Bipartitions and Beatty Sequences

Spring 2021 ARML Power Contest

The Solutions

1. The complement must contain all, and only those positive integers that are not in the original set. That would be the set of even positive integers, $\{2, 4, 6, 8, \ldots\}$.

2. Since the positive integer 1 is neither prime nor composite, it does not belong to either of the proposed sets. Since the union of the two sets is not the entire set of positive integers, the two sets do not form a bipartition.

3. Recall the fact that if a is an integer and x is any real number, then $\lfloor x+a \rfloor = \lfloor x \rfloor + a$.

   (a) Given that positive integer n is in X, there must be a positive integer k such that $\lfloor k \cdot \frac{p}{q} \rfloor = n$. Then $k+q$ is a positive integer and $\lfloor (k+q) \cdot \frac{p}{q} \rfloor = \lfloor k \cdot \frac{p}{q} + p \rfloor = \lfloor k \cdot \frac{p}{q} \rfloor + p = n + p$ so, by definition, $n+p$ is in X.

   (b) Since m is in X' there is no positive integer k so that $\lfloor k \cdot \frac{p}{q} \rfloor = m$. Put another way, there must be some positive integer j so that $j \cdot \frac{p}{q} < m$ but $(j+1) \cdot \frac{p}{q} \geq m+1$. But if k is an integer with $k \leq j+q$ then

   $$k \cdot \frac{p}{q} \leq (j+q) \cdot \frac{p}{q} = j \cdot \frac{p}{q} + p < m + p$$

   while if $k \geq j+q+1$ then

   $$k \cdot \frac{p}{q} \geq (j+1+q) \cdot \frac{p}{q} = (j+1) \cdot \frac{p}{q} + p \geq m + 1 + p.$$

   Taken together, these two inequalities show that there is no positive integer k so that $\lfloor k\frac{p}{q} \rfloor = m + p$, and thus by definition $m+p$ must be in X'.

4. Rewrite $\frac{p}{q}$ as $1 + \frac{p-q}{q}$. Then $\lfloor k \cdot \frac{p}{q} \rfloor = \lfloor k + k \cdot \frac{p-q}{q} \rfloor = k + \lfloor k \cdot \frac{p-q}{q} \rfloor$. Therefore $k \cdot \frac{p-q}{q} < 1$ implies that $\lfloor k \cdot \frac{p}{q} \rfloor = k$. Thus the least number *not* in X—the least number in X'—must be the least k such that $k \cdot \frac{p-q}{q} \geq 1$. For this k, solving the inequality for k shows it is the least integer such that $k \geq \frac{q}{p-q}$, which is the definition of $\lceil \frac{q}{p-q} \rceil$.

5. The first ten multiples of ϕ are (approximately)

   $$1.618, 3.236, 4.854, 6.472, 8.090, 9.708, 11.326, 12.944, 14.562, 16.180$$

   so the Beatty sequence generated by ϕ^2 is

   $$1, 3, 4, 6, 8, 9, 11, 12, 14, 16.$$

6. The first ten multiples of ϕ^2 are (approximately)

   $$2.618, 5.236, 7.854, 10.472, 13.090, 15.708, 18.326, 20.944, 23.562, 26.180$$

   so the Beatty sequence generated by ϕ^2 is

   $$2, 5, 7, 10, 13, 15, 18, 20, 23, 26.$$

7. Since $\phi^2 \neq 0$, divide the equation $\phi^2 = \phi + 1$ by ϕ^2 to obtain the desired result.

8. Assume for the sake of contradiction that there are positive integers m, i, and j such that $m = \lfloor i\alpha \rfloor = \lfloor j\beta \rfloor$. By definition $m \leq i\alpha < m+1$ and $m \leq j\beta < m+1$. Since α and β are both irrational, the inequalities are both actually strict. That is, $m < i\alpha < m+1$ and $m < j\beta < m+1$. Divide the first inequality by α and the second by β to obtain
$$\frac{m}{\alpha} < i < \frac{m+1}{\alpha} \text{ and } \frac{m}{\beta} < j < \frac{m+1}{\beta}.$$
Now add these inequalities:
$$m\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) < i + j < (m+1)\left(\frac{1}{\alpha} + \frac{1}{\beta}\right).$$
One of the conditions on α and β is that $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. So the inequality reduces to $m < i + j < m + 1$. Since m is an integer there are no other integers strictly between m and $m+1$, but since i and j are integers a contradiction has been reached.

9. Assume for the sake of contradiction that there is a positive integer m that is not in the Beatty sequence generated by α nor in the sequence generated by β. Thus, there must be positive integers i and j such that
$$i\alpha < m < m+1 \leq (i+1)\alpha \text{ and } j\beta < m < m+1 \leq (j+1)\beta.$$
As in the previous solution, all inequalities must actually be strict since both α and β are irrational. Dividing the first inequality by α and the second by β yields
$$i < \frac{m}{\alpha} < \frac{m+1}{\alpha} < i+1 \text{ and } j < \frac{m}{\beta} < \frac{m+1}{\beta} < j+1.$$
Adding these two inequalities:
$$i + j < m\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) < (m+1)\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) < i + j + 2.$$
Invoke the constraint that $\frac{1}{\alpha} + \frac{1}{\beta} = 1$ to obtain
$$i + j < m < m + 1 < i + j + 2.$$
This implies that there are two integers strictly between the integers $i + j$ and $i + j + 2$ which is, of course, impossible.

10. The author admits this problem was much harder than it was supposed to be. The author had a solution in mind, but hadn't finished the details—which turned out to be impossible to finish! The steps of a solution are outlined below, with the details at the end of these solutions so as to not interrupt the flow of the document. Simpler solutions would be welcomed!

    The statement is true, and here is the outline of a proof:

- Given an irrational number x and a positive integer n, show that there are infinitely many positive integers k such that the fractional part of kx is between $\frac{i}{n}$ and $\frac{i+1}{n}$ for each $i = 0, 1, 2, \ldots n-1$. The fractional part of kx is defined to be $kx - \lfloor kx \rfloor$.
- Given an irrational number $\beta > 0$ and any arithmetic sequence $\{a_0 + jd\}_{j=1}^{\infty}$ with a_0 and d positive integers, there is some multiple $k\beta$ such that $\lfloor k\beta \rfloor$ is a member of this arithmetic sequence.
- Given that the Beatty sequence for an irrational number $\alpha > 1$ contains all the terms of the arithmetic sequence $\{a_0 + jd\}_{j=1}^{\infty}$ then the complementary set to this sequence contains *no* members of this sequence. But since the complementary set is the Beatty sequence generated by β where $\frac{1}{\alpha} + \frac{1}{\beta} = 1$ this would contradict the conclusion of the previous bullet.

11. This is a straightforward calculation: $\lfloor n\phi^2 \rfloor - \lfloor n\phi \rfloor = \lfloor n(\phi + 1) \rfloor - \lfloor n\phi \rfloor = \lfloor n\phi + n \rfloor - \lfloor n\phi \rfloor$. Since n is an integer, this can be rewritten as $\lfloor n\phi \rfloor + n - \lfloor n\phi \rfloor = n$.

12. (a) Consider the proof in problem 8. There, the reason that no number could occur in both Beatty sequences was that there were no integers strictly between m and $m+1$. However, since x and y are both rational the inequalities in that proof are no longer strict, and a positive m can appear in both X and Y as long as $m = ix = jy$ for some i and j. Since $x = \frac{p}{q}$ is given in lowest terms, $ix = i\frac{p}{q}$ is an integer if and only if $i = kq$ for some k, in which case $ix = kp$. Similarly, jy can only be an integer if $j = k(p-q)$ so that $jy = kp$. Since these choices of i and j clearly cause every multiple kp to be a member of both X and Y, the desired conclusion obtains.

    (b) As in the previous part, using $i = kq$ gives $ix = kp$ so $\lfloor ix \rfloor = kp$. But since $x > 1$, the previous multiple $(i-1)x = kp - x < kp - 1$ so $\lfloor (i-1)x \rfloor < kp - 1$. So no number of the form $kp - 1$ can be in X, and the proof that no such number can be a member of Y is similar. The proof that these are the only numbers not in either X or Y is similar to the proof of problem 9, which applies here *unless* $m+1$ is an integer that is an integer multiple of both x and y. Since the only such numbers are multiples of p, every positive integer m is a member of either X or Y unless $m + 1 = kp$, that is, unless $m \in \{kp - 1\}_{k=1}^{\infty}$.

13. Let $n \in Y_\alpha$. Then $\lfloor (n+1)\alpha \rfloor = \lfloor n\alpha \rfloor + 1$. Since α is irrational, there must be an integer, k, strictly between $n\alpha$ and $(n+1)\alpha$. Since $\alpha < 1$ there can only be one such k. So $n\alpha < k < (n+1)\alpha$. Thus $n < \frac{k}{\alpha} < n+1$, so $n = \lfloor k\frac{1}{\alpha} \rfloor$ and n is in the Beatty sequence generated by $\frac{1}{\alpha}$. Conversely, if n is in the Beatty sequence generated by $\frac{1}{\alpha}$ then there must be a k so that $\lfloor k\frac{1}{\alpha} \rfloor = n$, or $n \leq \frac{k}{\alpha} < n+1$. Since α is irrational the inequality is actually strict. Multiplying by α yields $n\alpha < k < (n+1)\alpha$. Recalling that $0 < \alpha < 1$ yields $k - 1 < k - \alpha < n\alpha < k < (n+1)\alpha < k + \alpha < k + 1$ so $\lfloor n\alpha \rfloor = k - 1$ while $\lfloor (n+1)\alpha \rfloor = k = \lfloor n\alpha \rfloor + 1$ so $n \in Y_\alpha$.

14. First it should be noted that Q^* is itself a nondecreasing sequence of nonnegative integers, so the sequence Q^{**} is a well-defined sequence. Let $q_n = k$ be the nth term

of the sequence Q. Since Q is nondecreasing there are at least n terms in Q that are equal to k or less, but at most $n-1$ terms that are less than k. So $f_k \leq n-1$ while $f_{k+1} \geq n$. Thus there are exactly k terms in Q^* that are less than n so that the nth term of Q^{**} is $k = q_n$.

15. Since all terms of Q are nonnegative, add a positive integer to a term makes it positive. And since Q is nondecreasing, $q_i \leq q_{i+1}$ so $q_i + i \leq q_{i+1} + i < i_{i+1} + i + 1$, rendering P strictly increasing.

16. Each positive integer n must be shown to be in P or in P^*, but not both. The proof will be by induction. Keep in mind that both Q and Q^* are nondecreasing sequences of nonnegative integers, so by problem 15 both P and P^* are strictly increasing sequences of positive integers.

    For the base case $n = 1$ consider two possibilities: $q_1 = 0$ or $q_1 > 0$.

    - If $q_1 = 0$ then $p_1 = q_1 + 1 = 1$ so $1 \in P$. On the other hand, $f_1 \geq 1$ so $p_1^* \geq 1 + 1 = 2$ and $1 \notin P^*$.
    - If $q_1 > 0$ then $p_1 = q_1 + 1 > 1$ and $1 \notin P$. But now $f_1 = 0$ so $p_1^* = f_1 + 1 = 1$.

    In either case, $n = 1$ is in either P or P^*, but not both.

    For the inductive step, assume without loss of generality that $n \in P$ but $n \notin P^*$. No generality is lost because if $n \in P^*$ but $n \notin P^*$ we appeal to problem 14 to reverse the roles of Q and Q^* and of P and P^*.

    Since $n \in P$ there is a unique k so that $n = p_k = q_k + k$. Let $r = n - k = q_k$. Again, consider two possibilities: $q_{k+1} = r$ or $q_{k+1} > r$. Each will be examined in turn.

    If $q_{k+1} = r$ then $p_{k+1} = r + k + 1 = n + 1$ so $n + 1 \in P$. On the other hand since $q_k = r$ and Q is nondecreasing, there are at most $k - 1$ terms of Q that are less than r, so $f_r < k$ and $p_r^* = f_r + r < k + r = n$. But since $q_{k+1} = r$ there are at least $k+1$ terms of Q that are less than $r+1$, so $f_{r+1} \geq k+1$ and $p_{r+1}^* \geq (k+1)+(r+1) = (k+r)+2 = n+2$ Since P^* is strictly increasing, $p_r^* < n$ and $p_{r+1}^* > n + 1$, $n + 1 \notin P^*$. So when both q_k and q_{k+1} are equal to r, both n and $n + 1$ are in P and neither is in P^*.

    The other possibility is that $q_{k+1} > r$. That is, since q_{k+1} is an integer, $q_{k+1} \geq r + 1$ Then $p_{k+1} > r + k + 1 > n + 1$ so $n + 1 \notin P$. On the other hand, $f_r < k$ as before, but $f_{r+1} = k$ because q_k is the last term in Q that is less than $r + 1$. Since $f_{r+1} = k$, by definition $p_{r+1}^* = k + r + 1 = n + 1$ so $n + 1 \in P^*$. Thus, in this situation $n + 1$ is in exactly one of P or P^*.

    These two cases cover all possibilities, so the induction step is complete.

Finally, here are two lemmas which complete the proof outlined in problem 10.

Lemma: Given an irrational number x and a positive integer n, there are infinitely many positive integers k such that the fractional part of kx is between $\frac{i}{n}$ and $\frac{i+1}{n}$ for each $i = 0, 1, 2, \ldots n-1$. The fractional part of kx is defined to be $kx - \lfloor kx \rfloor$.

Proof: Assume there is some i for which there are only finitely many such k. Then there is a largest such, which will be denoted K. Consider the fractional parts of $(K+1)x$, $(K+2)x$, $\ldots$, $(K+n)x$. None of them are between $\frac{i}{n}$ and $\frac{i+1}{n}$, so by the pigeonhole principle there is some other of these intervals of width $\frac{1}{n}$ that contains at least two of these fractional values. Those two fractional values differ by less than $\frac{1}{n}$. In other words, there are a and b with $1 \leq a < b \leq n$ such that $(K+b)x - \lfloor (K+b)x \rfloor - ((K+a)x - \lfloor (K+a)x \rfloor) < \frac{1}{n}$. Thus the fractional part of $(b-a)x$ is between $-\frac{1}{n}$ and $\frac{1}{n}$ (and is not zero!).

Adding multiples of $(b-a)x$ to Kx now adds a fraction less than $\frac{1}{n}$ to the fractional part, so some one of these multiple of x must have a fractional part between $\frac{i}{n}$ and $\frac{i+1}{n}$. ■

Lemma: Given an irrational number $\beta > 0$ and any arithmetic sequence $\{a_0 + jd\}_{j=1}^{\infty}$ with a_0 and d positive integers, there is some multiple $k\beta$ such that $\lfloor k\beta \rfloor$ is a member of this arithmetic sequence.

Proof: Let $x = \beta/d$ and let $i \equiv a_0 \pmod{d}$ with $0 \leq i < d$. By the previous lemma there are infinitely many choices of k so that the fractional part of kx is between $\frac{i}{d}$ and $\frac{i+1}{d}$. So for each of these k,

$$\begin{gathered}
\frac{i}{d} < kx - \lfloor kx \rfloor < \frac{i+1}{d} \\
i < dkx - d\lfloor kx \rfloor < i+1 \\
i < k\beta - d\lfloor kx \rfloor < i+1 \\
i + d\lfloor kx \rfloor < k\beta < i + 1 + d\lfloor kx \rfloor \\
\lfloor k\beta \rfloor = i + d\lfloor kx \rfloor
\end{gathered}$$

Now $\lfloor kx \rfloor$ is an integer, so $d\lfloor kx \rfloor$ is a multiple nd of d. Since k can be chosen arbitrarily large, n can be arbitrarily large, so eventually $nd > a_0$. Then $i + nd$ is a member of the arithmetic sequence $\{a_0 + jd\}_{j=1}^{\infty}$. ■

Beatty sequences are names for Samuel Beatty, who was the first Canadian citizen to earn a Ph.D. in mathematics. He became a student at the University of Toronto in 1903 and stayed there for his entire life, earning all his degrees there, becoming a professor, and eventually the university's chancellor. He was the only doctoral student of John Charles Fields, after whom the Fields medal—the "Nobel Prize of mathematics"—is named. In 1926 he published a problem in the *American Mathematical Monthly* which was essentially the theorem about complementary Beatty sequences. Lord Rayleigh had actually already solved this problem almost 50 years earlier, but almost no one had seen that work. Most historians of math problems consider this problem to be the source of more research than any other problem ever published in the problems section of a journal.